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Quantum Attention

Preprint · September 2025

DOI: 10.13140/RG.2.2.19514.25289

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Gary Nan Tie, Sep 19, 2025

In category theory, perspective and context are everything,
and so it is with transformer attention.

Introduction

In quantum mechanics, collapse of states occurs when a measurement is made, forcing a quantum system to transition from a superposition of multiple possible states to a single definite state. This process is essential for connecting quantum mechanics with classical observations and is described by various interpretations, including the Copenhagen interpretation and many-worlds interpretation. This phenomenon is closely related to quantum entanglement, where the states of entangled particles are interdependent, meaning the measurement of one particle instantly affects the state of others.

Curiously, a similar phenomenon occurs in fibration attention, in the following sense. Consider an effect that can explain an observation by multiple plausible hypotheses. Abductive inference works in reverse, choosing a single hypothesis whose effect is the given observation. A key-value pairing $p: K \rightarrow V$, is an effect. We say a key k , explains value v , if the effect of k is v , that is $p(k) = v$. The inverse image of v under p , unless one-to-one, typically has multiple meanings k , providing semantics for v . A section $\tau: V \rightarrow K$ of effect $p: K \rightarrow V$, that is satisfies $p(\tau(v)) = v$, is precisely abduction, of multiple potential explanations, it is choosing one. Note that this is form state/key collapse.

Moreover, a query is a similarity measurement on a system of keys determining the meaning of a value. In fibration attention, the entanglement of measurements and collapses from a sequence of queries, is characterized by a fibration lifting. A lifting is a neural network that simultaneously learns how to update an input query and to abductively find a key that explains the value being output. This construction is one of the foundational building blocks for fibration attention architectures.

So abduction is state collapse post measurement and fibration lifting is entanglement. What is new here is recognizing the quantum nature of fibration attention; a new paradigm for transformer attention beyond database retrieval, let's call this perspective 'Quantum Attention'.

For completeness, we now describe in detail, attention constructed by fibration liftings. Various architectures and application have been explored in prior work. Fortunately, quantum computers are not needed to implement quantum attention!

Notation

Tokens representing words are projected onto a lower dimensional latent space $\mathcal{X} \subseteq \mathbb{R}^d$.

In $\mathcal{X}$, let subsets $\mathcal{Q}$, $\mathcal{K}$, and $\mathcal{V}$, denote queries, keys and values respectively, and let $\mathcal{Q}'$ a copy of $\mathcal{Q}$, denote updated queries.

Data samples to learn neural networks from:

$$q_i \in \mathcal{Q}, \quad i \in [n] \triangleq \{1, 2, \dots, n\}$$

$$(k_j, v_j) \in \mathcal{K} \times \mathcal{V}, \quad j \in [m]$$

Let $L: \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^+$ be a loss function.

For $\varepsilon > 0$, define $x \approx_\varepsilon y$ iff $L(x, y) < \varepsilon$.

Suppose there exists $\varepsilon > 0$ such that we can learn a neural network $p: K \rightarrow V$ with $p(k_j) =_{\varepsilon} v_j$, $j \in [m]$.

We will use effect p , a key-value semantic pairing, to construct a weak factorization system fibration that characterizes entanglement when a similarity measurement, $m: Q \rightarrow Q'$, a query update, is made.

Transformer attention via fibration

In a category with neural network arrows,
consider the following Joyal-Tierney
symbolic calculus diagram, commuting up to $\approx_\varepsilon$.

$$\begin{array}{ccc}
 & m \otimes 1 & \\
 Q \otimes K & \xrightarrow{\quad} & Q' \otimes K \\
 \downarrow 1 \otimes p & \nearrow \sigma & \downarrow 1 \otimes p \\
 Q \otimes V & \xrightarrow{m \otimes 1} & Q' \otimes V
 \end{array} \quad (†)$$

Learn lifting NN $\sigma: Q \otimes V \rightarrow Q' \otimes K$ such that:

$$i) \quad \sigma \begin{bmatrix} q_i \\ p(k_j) \end{bmatrix} \approx_\varepsilon \begin{bmatrix} m(q_i) \\ k_j \end{bmatrix}, \quad i \in [n], j \in [m]$$

$$ii) \quad \pi_1 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} \approx_\varepsilon m(q_i), \quad i \in [n], j \in [m]$$

$$iii) \quad p \pi_2 \sigma \begin{bmatrix} q_i \\ v_j \end{bmatrix} \approx_\varepsilon v_j, \quad i \in [n], j \in [m]$$

For $i \in [n]$, define $\iota_i : V \rightarrow Q \otimes V$

$$v \mapsto \begin{bmatrix} q_i \\ v \end{bmatrix}$$

and define $\tau_i : V \rightarrow K$ by $\tau_i \triangleq \pi_2 \sigma \iota_i$,

then $p(\tau_i(v_j)) =_{\varepsilon} v_j$, $j \in [m]$,

ie lifting σ determines sections τ_i of p .

—

Query updating

Given $q'_i \in Q'$, $i \in [n]$, learn NN

$m : Q \rightarrow Q'$ such that $m(q_i) =_{\varepsilon} q'_i$

Given $p : K \rightarrow V$ and $m : Q \rightarrow Q'$,

learn $\sigma^1 : Q \otimes V \rightarrow Q' \otimes K$ such that $(\dagger)$

commutes up to $=_{\varepsilon}$.

Let $\alpha_{ij} \triangleq \text{softmax score}(q'_i, k_j)$

and $j^* \triangleq \arg \max_j \alpha_{ij}$, where score

is a similarity function e.g. a reproducing kernel.

Define Fibration update $\vec{q}_i \triangleq \pi_1 \sigma^{-1} \begin{bmatrix} q'_i \\ v_{j^*} \end{bmatrix} \in Q'$

Finally, let $q''_i \triangleq \sum_{j=1}^m \beta_{ij} v_j$,

where $\beta_{ij} \triangleq \text{softmax}_j \text{ score}(\vec{q}_i, k_j)$.

This is a single update step from q'_i to q''_i .

Query update process: Replace q'_i with q''_i ,

to learn σ^2 and sections τ_i^2 , and so on.

Under Lyapunov conditions, $\tau_i^1(v_j)$, $\tau_i^2(v_j)$, ...

are refined explanation of v_j , in response to query q_i .

Initial measurement

Query updating began with given $q'_i \in Q'$, $i \in [n]$.

$$\text{Define } q'_i \triangleq \sum_{j=1}^m \gamma_{ij} v_j,$$

$$\text{where } \gamma_{ij} \triangleq \underset{j}{\text{softmax score}}(q'_i, k_j),$$

$$0 \leq \gamma_{ij} \leq 1.$$

Value returned to a query

If $q''_i = \sum_{j=1}^m \beta_{ij} v_j$ is the final update after successive

iterations, then v_{j^*} is returned in answer to

query q_i , where $j^* \triangleq \arg \max_j \beta_{ij}$.

Summary

A measurement $m(q_i)$ collapses the set of keys $p^{-1}(v_j)$, that explain v_j , to a single key $T_i(v_j)$.

Fibration lifting σ characterizes their entanglement.

Under Lyapunov conditions, successive measurement,

query updates, refine the meaning of v_j^*

under the semantics p .

Quantum attention, so to speak!



Reference

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DOI: 10.13140/RG.2.2.36796.09600